

Shijun ZHANG

Assistant Professor

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Appointments

Assistant Professor (Tenure-Track), Hong Kong Polytechnic University (PolyU), Hong Kong SAR, China	Jul 2024 ~ Present
Phillip Griffiths Assistant Research Professor , Duke University, United States Mentors: Jianfeng Lu and Hongkai Zhao	Aug 2022 ~ Jun 2024
Research Fellow , National University of Singapore, Singapore Mentor: Zuowei Shen	Jan 2021 ~ Jul 2022

Education

Ph.D. in Mathematics , National University of Singapore, Singapore Thesis: <i>Deep neural network approximation via function compositions</i> [PDF, URL] Supervisors: Zuowei Shen and Haizhao Yang	Aug 2016 ~ Jan 2021
B.S. in Mathematics , Wuhan University, China Thesis supervisor: Xiliang Lv	Sep 2012 ~ Jun 2016

Awards

The 2026 Frontiers of Science Award (Mathematics), International Congress for Basic Science (ICBS 2026), Award Medal, URL.

Departmental Best Paper Award, AMA, PolyU, 2024-2025, Award Trophy.

The EASIAM (East Asia section of SIAM) Student Paper Prize (First Prize), 2020-2021, URL.

Teaching

Will teach **Learning Theory for AI** (AMA 579) in 2026/27 Sem2.

Operations Research Methods (AMA 3820), PolyU 2025/26 & 2026/27 Sem1 & Sem2
Instructor, Syllabus

Optimization Methods (AMA 4850), PolyU 2024/25 & 2025/26 Sem2
Instructor, Syllabus

Mathematical Numerical Analysis (Math 361S), Duke University Spring 2024
Instructor, Syllabus

Matrices and Vectors (Math 218D-2), Duke University Fall 2023
Teaching assistant

Publications

[number] Author(s). *Paper title*. Journal or conference reference. [Links]

* Corresponding author † Equal contribution

Preprint(s)

- [23] Baicheng Li, Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Optimal tradeoffs between network size and parameter magnitude in neural approximation and minimax regression*. Submitted. [arXiv]
- [22] **Shijun Zhang**. *Sharp approximation rates for neural networks with affine latent parameterizations*. Submitted. [arXiv]
- [21] Baicheng Li, Haizhao Yang, **Shijun Zhang**. *Sobolev approximation by fixed-size neural networks with arbitrary accuracy*. Submitted. [arXiv]
- [20] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Layer-wise geometric approximation rates for deep networks*. Submitted. [arXiv]
- [19] Zeyu Li, **Shijun Zhang**, Tiejong Zeng, Feng-Lei Fan. *Neural network approximation: a view from polytope decomposition*. Submitted. [arXiv]
- [18] **Shijun Zhang***, Zuowei Shen, Yuesheng Xu. *Multigrade neural network approximation*. Submitted. [arXiv]
- [17] Hanyu Pei, Jing-Xiao Liao, Qibin Zhao, Ting Gao, **Shijun Zhang**, Xiaoge Zhang, Feng-Lei Fan. *Neuronseek: on stability and expressivity of task-driven neurons*. Submitted. [arXiv]

Published (Accepted)

- [16] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Fourier multi-component and multi-layer neural networks: unlocking high-frequency potential*. *Neural Networks*, 204:109268, 2026. [arXiv, Journal]
- [15] Fenglei Fan, Juntong Fan, Dayang Wang, Jingbo Zhang, Zelin Dong, **Shijun Zhang**, Ge Wang, Tiejong Zeng. *Hyper-compression: model compression via hyperfunction*. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 48(4):8813–8830, 2026. [arXiv, Journal]
- [14] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Structured and balanced multi-component and multi-layer neural networks*. *SIAM Journal on Scientific Computing*, 47(5):C1059-C1090, 2025. [arXiv, Journal]
- [13] **Shijun Zhang***, Hongkai Zhao, Yimin Zhong, Haomin Zhou. *Why shallow networks struggle to approximate and learn high frequencies*. *Information and Inference: A journal of the IMA*, 14(3):iaaf022, September 2025. [arXiv, Journal]
- [12] Qianchao Wang[†], **Shijun Zhang**[†], Dong Zeng, Zhaoheng Xie, Hengtao Guo, Feng-Lei Fan, Tiejong Zeng. *Don't fear peculiar activation functions: euaf and beyond*. *Neural Networks*, 186:107258, June 2025. [arXiv, Journal]
- [11] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *Deep network approximation: beyond ReLU to diverse activation functions*. *Journal of Machine Learning Research*, 25(35):1–39, 2024. [arXiv, Journal]
- [10] **Shijun Zhang***, Jianfeng Lu, Hongkai Zhao. *On enhancing expressive power via compositions of single fixed-size ReLU network*. *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202:41452–41487, 2023. [arXiv, Poster, Conference]

- [9] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Neural network architecture beyond width and depth*. Advances in Neural Information Processing Systems (**NeurIPS 2022**), 35:5669–5681, 2022. [arXiv, Poster, Conference]
- [8] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation in terms of intrinsic parameters*. Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**), PMLR 162:19909–19934, 2022. [arXiv, Spotlight, Conference]
- [7] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Deep network approximation: achieving arbitrary accuracy with fixed number of neurons*. **Journal of Machine Learning Research**, 23(276):1–60, September 2022. [arXiv, Journal]
- [6] Zuowei Shen, Haizhao Yang, **Shijun Zhang***. *Optimal approximation rate of ReLU networks in terms of width and depth*. **Journal de Mathématiques Pures et Appliquées**, 157:101–135, January 2022. [arXiv, Journal]
- [5] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *neural network approximation: three hidden layers are enough*. **Neural Networks**, 141:160–173, September 2021. [arXiv, Journal]
- [4] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network with approximation error being reciprocal of width to power of square root of depth*. **Neural Computation**, 33(4):1005–1036, April 2021. [arXiv, Journal]
- [3] Jianfeng Lu, Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation for smooth functions*. **SIAM Journal on Mathematical Analysis**, 53(5):5465–5506, September 2021. [arXiv, Journal]
- [2] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Deep network approximation characterized by number of neurons*. **Communications in Computational Physics**, 28(5):1768–1811, November 2020. [arXiv, Journal]
- [1] Zuowei Shen, Haizhao Yang, **Shijun Zhang**. *Nonlinear approximation via compositions*. **Neural Networks**, 119:74–84, November 2019. [arXiv, Journal]